Partial Differential Equations

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1 Introduction

1.1 Basic Concepts

Definition 1.1. *(Partial Differential Equation)*
Let $x = (x_1, \cdots, x_n)$ and let $u = u(x)$ be the unknown function. Write

$$Du = \left(u_{x_1}, \cdots, u_{x_n}\right), \quad D^k u = \left(\frac{\partial^k u}{\partial x_1^{k_1} \cdots \partial x_n^{k_n}}\right)_{k_1 + \cdots + k_n = k}.$$

An equation

$$F(x, u, Du, D^2 u, \cdots, D^N u) = f(x)$$

is a partial differential equation of order N, denoted by $Gu = f(x)$.

Definition 1.2. *(Multi-index)*
A multi-index is $\alpha = (\alpha_1, \cdots, \alpha_n)$ with length $|\alpha| = \alpha_1 + \cdots + \alpha_n$ and

$$\partial^\alpha = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}.$$

Definition 1.3. *(Homogeneous Equation)*
The equation $Gu = f(x)$ is homogeneous when $f \equiv 0$, and nonhomogeneous when $f \neq 0$.

Definition 1.4. *(Linear Equation)*
The equation is linear when $G(au+bv)=aGu+bGv$ for constants a,b. In multi-index form,

$$\sum_{|\alpha|\leq N} A_\alpha(x)\partial^\alpha u = f(x).$$

Otherwise the equation is nonlinear.

Definition 1.5. *(Semilinear, Quasilinear, and Fully Nonlinear Equations)*
The principal part is the collection of terms that contain a derivative of order N.

- The equation is semilinear when the principal part is linear, with coefficients depending only on x:

$$\sum_{|\alpha|=N} A_\alpha(x)\partial^\alpha u + A_0(x,u,Du,\cdots,D^{N-1}u) = f(x).$$

- The equation is quasilinear when it is linear in the derivatives of order N, whose coefficients may depend on derivatives of lower order:

$$\sum_{|\alpha|=N} A_\alpha(x,u,Du,\cdots,D^{N-1}u)\partial^\alpha u + A_0(x,u,Du,\cdots,D^{N-1}u) = f(x).$$

- The equation is fully nonlinear when a derivative of order N enters nonlinearly.

1.2 Typical Equations

Definition 1.6. *(The Wave Equation)*
Let $a>0$ and let $\Delta=\partial_{x_1x_1}+\cdots+\partial_{x_nx_n}$. The wave equation is

$$u_{tt}-a^2\Delta u = f$$

with initial conditions

$$u(x,0)=\varphi(x),\quad u_t(x,0)=\psi(x).$$

Definition 1.7. *(The Heat Equation)*
Let $a>0$. The heat equation, also called the diffusion equation, is

$$u_t-a^2\Delta u = f$$

with initial conditions

$$u(x,0)=\varphi(x).$$

Definition 1.8. *(Laplace's Equation)*
Laplace's equation is $\Delta u = 0$, and a solution of Laplace's equation is harmonic. The equation does not involve time, so there is no initial condition.

Definition 1.9. *(Boundary Conditions)*
Let $\Gamma = \partial\Omega$, let $\boldsymbol{n}$ be the outward unit normal, and let $\sigma > 0$. Dirichlet, Neumann, and Robin conditions are respectively given by

$$u|_\Gamma = g, \quad \left.\frac{\partial u}{\partial \boldsymbol{n}}\right|_\Gamma = g, \quad \left.\left(\frac{\partial u}{\partial \boldsymbol{n}} + \sigma u\right)\right|_\Gamma = g.$$

1.3 Second-Order Linear Equations

Definition 1.10. *(Discriminant and Type)*
Let $a_{11}^2 + a_{12}^2 + a_{22}^2 \neq 0$, and let $u = u(x, y)$ satisfy

$$a_{11}u_{xx} + 2a_{12}u_{xy} + a_{22}u_{yy} + F(x, y, u, Du) = 0.$$

The discriminant is $\Delta = a_{12}^2 - a_{11}a_{22}$, where
- $\Delta > 0$: the equation is hyperbolic;
- $\Delta = 0$: the equation is parabolic;
- $\Delta < 0$: the equation is elliptic.

Remark.
From here on the equation is linear,

$$Lu = a_{11}u_{xx} + 2a_{12}u_{xy} + a_{22}u_{yy} + b_1u_x + b_2u_y + cu = f,$$

with coefficients depending on (x, y). When a change of independent variables $\xi = \xi(x, y), \eta = \eta(x, y)$ is invertible near (x_0, y_0), the chain rule gives

$$u_x = u_\xi\xi_x + u_\eta\eta_x, \quad u_y = u_\xi\xi_y + u_\eta\eta_y,$$

$$u_{xx} = u_{\xi\xi}\xi_x^2 + 2u_{\xi\eta}\xi_x\eta_x + u_{\eta\eta}\eta_x^2 + u_\xi\xi_{xx} + u_\eta\eta_{xx},$$

$$u_{xy} = u_{\xi\xi}\xi_x\xi_y + u_{\xi\eta}\big(\xi_x\eta_y + \xi_y\eta_x\big) + u_{\eta\eta}\eta_x\eta_y + u_\xi\xi_{xy} + u_\eta\eta_{xy},$$

$$u_{yy} = u_{\xi\xi}\xi_y^2 + 2u_{\xi\eta}\xi_y\eta_y + u_{\eta\eta}\eta_y^2 + u_\xi\xi_{yy} + u_\eta\eta_{yy}.$$

Substitution produces

$$\overline{a}_{11}u_{\xi\xi} + 2\overline{a}_{12}u_{\xi\eta} + \overline{a}_{22}u_{\eta\eta} + \overline{b}_1 u_\xi + \overline{b}_2 u_\eta + \overline{c}u = \overline{f},$$

with $\overline{c} = c$, $\overline{f} = f$, and

$$\overline{a}_{11} = a_{11}\xi_x^2 + 2a_{12}\xi_x\xi_y + a_{22}\xi_y^2,$$

$$\overline{a}_{12} = a_{11}\xi_x\eta_x + a_{12}\big(\xi_x\eta_y + \xi_y\eta_x\big) + a_{22}\xi_y\eta_y,$$

$$\overline{a}_{22} = a_{11}\eta_x^2 + 2a_{12}\eta_x\eta_y + a_{22}\eta_y^2,$$

$$\overline{b}_1 = a_{11}\xi_{xx} + 2a_{12}\xi_{xy} + a_{22}\xi_{yy} + b_1\xi_x + b_2\xi_y,$$

$$\overline{b}_2 = a_{11}\eta_{xx} + 2a_{12}\eta_{xy} + a_{22}\eta_{yy} + b_1\eta_x + b_2\eta_y.$$

There is $\overline{\Delta} = |J|^2\ \Delta$ with $J = \begin{pmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{pmatrix}$, so the type is invariant.

Lemma 1.1.
If φ solves the principal equation above and $\varphi_x^2 + \varphi_y^2 \neq 0$, then

$$a_{11}(\mathrm{d}y)^2 - 2a_{12}\,\mathrm{d}x\,\mathrm{d}y + a_{22}(\mathrm{d}x)^2 = 0.$$

Proposition 1.1. *(Canonical Forms for Constant Coefficients)*
Suppose a_{ij}, b_j, and c are constant. A linear change of (x, y) puts the equation in one of the following forms, with constant coefficients on the right-hand side.

- Hyperbolic (first canonical form): $u_{\xi\eta} = Au_\xi + Bu_\eta + Cu + D$.
- Hyperbolic (second canonical form): $u_{rr} - u_{ss} = Au_r + Bu_s + Cu + D$.
- Parabolic: $u_{\eta\eta} = Au_\xi + Bu_\eta + Cu + D$.
- Elliptic: $u_{rr} + u_{ss} = Au_r + Bu_s + Cu + D$.

Example.
The equation $4u_{xx} + 5u_{xy} + u_{yy} + u_x + u_y = 2$ has $\Delta = \left(\frac{5}{2}\right)^2 - 4 > 0$, so it is hyperbolic. The characteristic equation $4(\mathrm{d}y)^2 - 5\,\mathrm{d}x\,\mathrm{d}y + (\mathrm{d}x)^2 = 0$ factors as $(4\,\mathrm{d}y - \mathrm{d}x)(\mathrm{d}y - \mathrm{d}x) = 0$, with characteristic lines $4y - x = C_1$ and $y - x = C_2$. Set $\xi = 4y - x$ and $\eta = y - x$. Then $\overline{a}_{11} = \overline{a}_{22} = 0$, $\overline{a}_{12} = -\frac{9}{2}$, $\overline{b}_1 = 3$, and $\overline{b}_2 = 0$, so $-9u_{\xi\eta} + 3u_\xi = 2$, that is

$$u_{\xi\eta} = \frac{1}{3}u_\xi - \frac{2}{9}.$$

Writing $w = u_\xi$ gives $w_\eta - \frac{w}{3} = -\frac{2}{9}$. Multiplication by $e^{-\frac{\eta}{3}}$ produces $w = \frac{2}{3} + F(\xi)e^{\frac{\eta}{3}}$, and one further integration gives

$$u(x,y) = \frac{2}{3}(4y - x) + f(4y - x)e^{\frac{y-x}{3}} + g(y - x).$$

Example.
The equation $u_{xx} - 4u_{xy} + 4u_{yy} = e^y$ has $\Delta = 2^2 - 4 = 0$, so it is parabolic. The characteristic equation $(\mathrm{d}y + 2\,\mathrm{d}x)^2 = 0$ has the lines $y + 2x = C$. Set $\xi = y + 2x$ and $\eta = y$. Then $\overline{a}_{11} = \overline{a}_{12} = 0$, $\overline{a}_{22} = 4$, and the lower-order coefficients coming from ξ and η vanish, so

$$u_{\eta\eta} = \frac{1}{4}e^\eta.$$

Two integrations give $u = \frac{1}{4}e^\eta + \eta f(\xi) + g(\xi)$, hence

$$u(x,y) = \frac{1}{4}e^y + yf(y + 2x) + g(y + 2x).$$

Example.
The equation $u_{xx} + u_{xy} + u_{yy} + u_x = 0$ has $a_{12} = \frac{1}{2}$ and $\Delta = \left(\frac{1}{2}\right)^2 - 1 < 0$, so it is elliptic. The characteristic equation $\mathrm{d}y^2 - \mathrm{d}x\,\mathrm{d}y + \mathrm{d}x^2 = 0$ has roots

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i.$$

Set $\xi = y - \frac{x}{2}$ and $\eta = -\frac{\sqrt{3}}{2}x$. Then

$$\overline{a}_{11} = \overline{a}_{22} = \frac{3}{4}, \quad \overline{a}_{12} = 0, \quad \overline{b}_1 = -\frac{1}{2}, \quad \overline{b}_2 = -\frac{\sqrt{3}}{2},$$

and the equation becomes

$$u_{\xi\xi} + u_{\eta\eta} = \frac{2}{3}u_\xi + \frac{2}{\sqrt{3}}u_\eta.$$

Example.
The equation $y^2u_{xx} - x^2u_{yy} = 0$ has $\Delta = x^2y^2$, so it is hyperbolic off the coordinate axes. The characteristic equation $y^2(\mathrm{d}y)^2 - x^2(\mathrm{d}x)^2 = 0$ factors as $(y\,\mathrm{d}y + x\,\mathrm{d}x)(y\,\mathrm{d}y - x\,\mathrm{d}x) = 0$, with integrals

$$\frac{1}{2}y^2 - \frac{1}{2}x^2 = C_1, \quad \frac{1}{2}y^2 + \frac{1}{2}x^2 = C_2.$$

Set $\xi = \frac{y^2-x^2}{2}$ and $\eta = \frac{y^2+x^2}{2}$. Then, wherever $|\xi| \neq |\eta|$,

$$u_{\xi\eta} = \frac{\eta}{2(\xi^2 - \eta^2)}u_\xi - \frac{\xi}{2(\xi^2 - \eta^2)}u_\eta.$$

Example.
The equation $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xyu_x + y^2u_y = 0$ has $\Delta = 0$, so it is parabolic. The characteristic equation $(x\,\mathrm{d}y - y\,\mathrm{d}x)^2 = 0$ has the lines $\frac{y}{x} = C$. Set $\xi = \frac{y}{x}$ and $\eta = y$. Then $u_{\eta\eta} = -u_\eta$. Writing $w = u_\eta$ and integrating twice gives

$$u(x,y) = -e^{-y}f\left(\frac{y}{x}\right) + g\left(\frac{y}{x}\right).$$

Example.
The equation $u_{xx} + x^2u_{yy} = 0$ has $\Delta = -x^2$, so it is elliptic off the axis $x = 0$. The characteristic equation $\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = -x^2$ has complex integrals $2y \pm ix^2 = C$. Set $\xi = 2y$ and $\eta = x^2$. Then, for $\eta \neq 0$,

$$u_{\xi\xi} + u_{\eta\eta} = -\frac{1}{2\eta}u_\eta.$$

Proposition 1.2. *(Reduced Canonical Forms)*
For constant coefficients, first-order terms may be removed by the exponential substitution $u = ve^{ar+bs}$. Starting from the first canonical forms and choosing the right a and b, this produces the reduced forms

$$v_{rs} = Ev + F, \quad v_{rr} \pm v_{ss} = Ev + F.$$

Definition 1.11. *(Second-Order Linear Operator)*
For $x = (x_1, \cdots, x_n)$ and a symmetric matrix $A = \left(a_{ij}\right)$,

$$Lu = \sum_{i,j=1}^{n} a_{ij}u_{x_ix_j} + \sum_{j=1}^{n} b_ju_{x_j} + cu = f.$$

The type at a point is read from the eigenvalues $\lambda_1, \cdots, \lambda_n$ of A.

Definition 1.12. *(Types in Several Variables)*
Let $\lambda_1, \cdots, \lambda_n$ be the eigenvalues of A.

- Hyperbolic: every λ_i is nonzero, $n-1$ of them have the same sign, and one has the opposite sign.
- Parabolic: exactly one eigenvalue is zero and the others have the same sign.
- Elliptic: every λ_i is nonzero and all have the same sign.
- Ultrahyperbolic: for $n \geq 4$, every λ_i is nonzero, at least two are positive, and at least two are negative.
- Ultraparabolic: more than one eigenvalue is zero and the remaining eigenvalues have the same sign.

1.4 Characteristics

Definition 1.13. *(Characteristic Direction)*
At a fixed point x, a direction $\boldsymbol{l} = (\alpha_1, \cdots, \alpha_n)$ is a characteristic direction if it satisfies the characteristic equation

$$\sum_{i,j=1}^{n} a_{ij}(x)\alpha_i\alpha_j = 0.$$

Definition 1.14. *(Characteristic Surface)*
The hypersurface $S : \varphi(x) = 0$ is a characteristic surface if at every point of S the normal $\left(\varphi_{x_1}, \cdots, \varphi_{x_n}\right)$ is a characteristic direction, namely

$$\sum_{i,j=1}^{n} a_{ij}(x)\varphi_{x_i}\varphi_{x_j} = 0.$$

Remark.
To find characteristic surfaces, solve the first-order equation above for φ. When $n = 2$, along $\varphi = C$ one has $\varphi_x\,\mathrm{d}x + \varphi_y\,\mathrm{d}y = 0$, hence $\mathrm{d}y/\,\mathrm{d}x = -\varphi_x/\varphi_y$. Substituting the normal $\left(\varphi_x, \varphi_y\right)$ into the characteristic equation gives the ODE

$$a_{11}(\mathrm{d}y)^2 - 2a_{12}\,\mathrm{d}x\,\mathrm{d}y + a_{22}(\mathrm{d}x)^2 = 0$$

used to reduce the two-variable equation to canonical form. In two variables a characteristic surface is a characteristic curve.

Definition 1.15. *(Characteristic Plane)*
A characteristic plane is an $(n-1)$-dimensional hyperplane whose normal is a characteristic direction. Through a point $\overline{x}$ with characteristic direction $(\alpha_1, \cdots, \alpha_n)$ it is

$$\sum_{i=1}^{n} \alpha_i (x_i - \overline{x}_i) = 0.$$

Definition 1.16. *(Characteristic Cone)*
The characteristic cone at a fixed point $\overline{x}$ is the cone formed by the envelope of all characteristic planes through $\overline{x}$.

Remark.
If the characteristic planes through $\overline{x}$ form a family $F(x, \lambda) = 0$, the envelope is obtained by eliminating the parameters λ from $F = 0$ and the equations obtained by differentiating F with respect to each parameter. Away from the vertex, the cone is itself a characteristic surface.

Example.
For Laplace's equation $\Delta u = 0$, the characteristic equation is $\alpha_1^2 + \cdots + \alpha_n^2 = 0$. There is no real characteristic direction, hence no real characteristic surface.

Example.
For the heat equation $u_t = a^2\big(u_{xx} + u_{yy} + u_{zz}\big)$, write $\boldsymbol{l} = (\alpha_0, \alpha_1, \alpha_2, \alpha_3)$ with $|\boldsymbol{l}| = 1$. The principal part involves only the spatial second derivatives, so the characteristic equation is $\alpha_1^2 + \alpha_2^2 + \alpha_3^2 = 0$. Together with $|\boldsymbol{l}| = 1$ this forces $\alpha_0 = \pm 1$ and $\alpha_1 = \alpha_2 = \alpha_3 = 0$. The characteristic directions are $\boldsymbol{l} = (\pm 1, 0, 0, 0)$, and the characteristic planes are $t = \text{const}$. Through a given point there is only one such plane, so the characteristic cone degenerates to that plane.

Example.
For the wave equation $u_{tt} = a^2\big(u_{xx} + u_{yy}\big)$, the characteristic equation $\alpha_0^2 = a^2(\alpha_1^2 + \alpha_2^2)$ together with $|\boldsymbol{l}| = 1$ gives

$$\alpha_0 = \pm \frac{a}{\sqrt{1 + a^2}}, \quad \alpha_1^2 + \alpha_2^2 = \frac{1}{1 + a^2}.$$

Parameterizing the spatial part yields the characteristic directions

$$\boldsymbol{l} = \left(\pm \frac{a}{\sqrt{1+a^2}}, \frac{\cos\varphi}{\sqrt{1+a^2}}, \frac{\sin\varphi}{\sqrt{1+a^2}} \right).$$

If θ is the angle between $\boldsymbol{l}$ and the t-axis, then $\tan\theta = 1/a$. Scaling the normal, the characteristic planes through (t_0, x_0, y_0) are

$$a(t-t_0) + \cos\varphi \cdot (x-x_0) + \sin\varphi \cdot (y-y_0) = 0.$$

Differentiating with respect to φ yields $-\sin\varphi \cdot (x-x_0) + \cos\varphi \cdot (y-y_0) = 0$. Squaring and adding these two equations eliminates φ and produces the characteristic cone

$$(x-x_0)^2 + (y-y_0)^2 = a^2 (t-t_0)^2.$$